

Bilateral Module — Technical Documentation

Server package `server/researchindicators` · NestJS 11 + TypeORM/MySQL · Epic AC-1385 (Pool-funding & ToC alignment)

Audience	ARI backend / technical team		
Scope	The <code>bilateral</code> entity module + the <code>toc-integration</code> tool module, as they exist in code today		
Branch	<code>AC-1594-bilateral-module-v2</code>		
Generated	2026-06-17		
Legend	NEW ToC-mapping-v2 (US2/AC-1594)	FOUNDATION earlier pool-funding work	MODIFIED

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1. Overview & architecture

The bilateral module lets a STAR contributor declare, per result, how a **pool-funding** result aligns with each Science Program's **Theory of Change (ToC)**. It is composed of two layers built in sequence:

Layer	What it provides	Tag
Foundation (pending-items / T-15.x)	Pool-funding eligibility & alignment persistence, SP-resolution chain (result → AGRESSO contract → <code>bilateral_project_mapping</code> → CLARISA project → SP codes), AGRESSO pool-funding tag, indicator-mapping with per-type handlers, review history, <code>ResultOwnerGuard</code> .	FOUNDATION
ToC Mapping v2 (US2 / AC-1594)	Live <code>lambda-toc</code> catalog read (per SP × level), server-owned <code>result_type → allowed_levels</code> rule, per-SP ToC alignment persistence with display snapshots, hardcoded live-version 2026 gate.	NEW

Request flow (ToC mapping v2)

```
BilateralController (/api/v1/results/:resultCode/pool-funding-alignment)
├ GET /hlos-indicators → BilateralService.getHlosIndicatorsForResult
│   ├── ResultRepository.findPoolFundingAlignmentContext (report_year_id, indicator_id, ...)
│   ├── toc-level-rules.util → result_type + allowed_levels [NEW]
│   ├── SP-resolution chain (AGRESSO → bilateral_project_mapping → CLARISA → SP)
│   └─ TocIntegrationService.getTocResultsForSps(spCodes, levels) [NEW lambda-toc]
├ GET / → getAlignment (+ version_locked, toc_alignments[]) [MODIFIED]
└ PATCH / → updateAlignment (+ toc_alignments[] upsert, gate, cascade) [MODIFIED]
    └─ ResultPoolFundingTocAlignmentRepository.upsertForSp / deactivateForSps [NEW]
```

Key principle: the level rule (`result_type → allowed_levels`) lives in *one* util used by both the read and the write paths, so the FE never re-derives it and read/write can never diverge.

2. Data model

Six tables. Five are persisted by this module; `results` is the parent (FK target). All extend `AuditableEntity` (`created_at/by` , `updated_at/by` , `is_active` , `deleted_at`). Engine InnoDB, charset `utf8mb4_unicode_520_ci` .

NEW `result_pool_funding_toc_alignment` `migration 1779190000015`

One row per (`result` , `SP`) capturing the per-SP ToC answer plus display **snapshots** (so saved alignments survive upstream catalog drift — R-BIL-095).

Column	Type	Notes
<code>id</code>	bigint PK auto	Primary key
<code>result_id</code>	bigint NOT NULL	FK → <code>results.result_id</code> (<code>fk_rpfta_result</code>)
<code>sp_code</code>	varchar(50) NOT NULL	Science Program code, e.g. <code>SP01</code>
<code>aligns_with_toc</code>	tinyint(1) NOT NULL	The per-SP Yes/No answer
<code>level</code>	varchar(10) NULL	<code>OUTPUT</code> / <code>OUTCOME</code> / <code>EOI</code> ; null on "No"
<code>toc_result_id</code>	int NULL	Chosen upstream ToC result
<code>indicator_id</code>	int NULL	Chosen upstream indicator
<code>quantitative_contribution</code>	decimal(18,2) NULL	Contributor's numeric value
<code>toc_result_title</code>	text NULL	Snapshot
<code>indicator_description</code>	text NULL	Snapshot
<code>unit_messurament</code>	varchar(100) NULL	Snapshot — upstream spelling kept verbatim (D-V2-4); exposed on the wire as <code>unit_of_measurement</code>
<code>target_value</code>	varchar(50) NULL	Snapshot (resolved for the target year)
<code>target_year</code>	int NULL	Snapshot — 2026
<code>active_result_sp</code>	varchar(71) STORED GENERATED	<code>IF(is_active=1, CONCAT(result_id,':',sp_code), NULL)</code>

Indexes: `UNIQUE idx_rpfta_active_result_sp(active_result_sp)` — enforces *one active row per (result, SP)*; inactive rows have NULL there and NULLs don't collide. `INDEX idx_rpfta_result(result_id)` . Flipping cardinality to N-per-SP later = drop one index, nothing else.

FOUNDATION `result_pool_funding_alignment` `1779190000006` & `..._sp` `1779190000007`

Column	Type	Notes
result_pool_funding_alignment		
<code>id</code>	bigint PK auto	
<code>result_id</code>	bigint NOT NULL	FK → <code>results</code> (<code>fk_rpfa_result</code>)
<code>has_contribution</code>	tinyint NOT NULL	Whether the result contributes
result_pool_funding_alignment_sp (child — the selected SPs / "levers")		
<code>alignment_id</code>	bigint NOT NULL	FK → alignment (<code>fk_rpfas_alignment</code>)
<code>lever_code</code>	varchar(50) NOT NULL	Selected SP code

Index: `UNIQUE uq_result_pool_funding_alignment_result_active(result_id, is_active)` — one active alignment per result. SP rows are managed deactivate-and-recreate.

FOUNDATION `result_pool_funding_indicator_mapping` `1779190000008`

The earlier indicator-mapping path (distinct from ToC alignment). Links a result+lever+indicator to a typed contribution row via one of four nullable FKs.

Column	Type	Notes
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<code>result_id / lever_code / indicator_code</code>	bigint / varchar(50) / varchar(100)	The mapped triple
<code>indicator_type</code>	varchar(50)	Drives which handler runs
<code>result_capacity_sharing_id</code>	bigint NULL	FK → <code>result_capacity_sharing</code>
<code>result_knowledge_product_id</code>	bigint NULL	FK → <code>result_knowledge_products</code>
<code>result_policy_change_id</code>	bigint NULL	FK → <code>result_policy_change</code>
<code>result_innovation_dev_id</code>	bigint NULL	FK → <code>result_innovation_dev</code>
<code>other_contribution_narrative</code>	text NULL	Free text
<code>is_stale</code>	tinyint DEFAULT 0	Marked when the source contribution changes

Index: `UNIQUE uq_rpfim_result_indicator_active(result_id, lever_code, indicator_code, is_active)`.

FOUNDATION `bilateral_project_mapping` `1779190000011` • `result_review_history` `1779190000009`

Column	Type	Notes
bilateral_project_mapping — maps an AGRESSO agreement to a CLARISA project		
<code>agresso_agreement_id</code>	varchar(50) NOT NULL	FK-by-value to <code>agresso_contract</code>
<code>clarisa_project_id</code>	int NOT NULL	Upstream CLARISA <code>project.id</code>
<code>clarisa_project_short_name</code>	varchar(500) NULL	Snapshot at mapping time (D-PI-11)
<code>active_agreement_id</code>	varchar(50) STORED GENERATED	<code>IF(is_active=1, agresso_agreement_id, NULL) → UNIQUE uk_bpm_active_agreement</code>
result_review_history — audit of review decisions & alignment changes		
<code>result_id / version_id / actor_user_id</code>	bigint	Who/what
<code>event_type / decision</code>	varchar(50) / varchar(20)	Event + outcome
<code>justification</code>	text NULL	+ JSON <code>payload_before/after</code> columns

Partial-unique pattern (used 3x): a `STORED GENERATED` column that is non-NULL only while `is_active=1`, plus a `UNIQUE` index on it. This enforces "one active row per key" while allowing unlimited soft-deleted history — MySQL treats NULLs as non-colliding. Used by `toc_alignment`, `bilateral_project_mapping`, and (variant) the alignment table.

3. File structure & responsibilities

Two modules. **NEW** = added by ToC mapping v2; **MOD** = modified by it.

Tool module — `src/domain/tools/toc-integration/` **NEW**

File	Responsibility
<code>toc-integration.service.ts</code>	lambda-toc HTTP client: per-(SP, level) GET, 5-min keyed cache, warm-stale / cold-503 resilience, parallel fan-out
<code>toc-integration.module.ts</code>	Singleton tool module (HttpModule import, service export)
<code>dto/toc-integration.types.ts</code>	Upstream payload types (<code>TocResult</code> , <code>TocIndicator</code> , <code>TocIndicatorTarget</code> , <code>TocLevel</code>) — mirror upstream verbatim incl. the <code>unit_messurament</code> typo
<code>toc-integration.service.spec.ts</code>	12 unit tests (cache TTL, warm/cold failure, empty payload, fan-out, missing host)

Entity module — `src/domain/entities/bilateral/`

File	Tag	Responsibility
<code>bilateral.controller.ts</code>	MOD	7 endpoints under <code>.../pool-funding-alignment</code> ; Swagger
<code>bilateral.service.ts</code> (1,417 lines)	MOD	All read/write logic, SP-resolution, validation, snapshots
<code>bilateral.module.ts</code>	MOD	Wiring; registers the 4 entities, 4 repos, 5 handlers
<code>utils/toc-level-rules.util.ts</code>	NEW	<code>resolveResultTypeKey</code> , <code>allowedLevelsFor</code> , <code>MAPPABLE_LIVE_VERSION = 2026</code>
<code>entities/result-pool-funding-toc-alignment.entity.ts</code>	NEW	ToC alignment entity (14 columns + audit)
<code>repositories/result-pool-funding-toc-alignment.repository.ts</code>	NEW	<code>findActiveByResultId</code> , <code>upsertForSp</code> , <code>deactivateForSps</code> (EntityManager-aware)
<code>dto/bilateral-hlos-indicators.response.dto.ts</code>	MOD	Frozen catalog-read envelope (6 response classes)
<code>dto/update-pool-funding-alignment.dto.ts</code>	MOD	+ <code>TocAlignmentInputDto</code> nested validation
<code>entities/...alignment.entity.ts</code> , <code>...alignment-sp</code> , <code>...indicator-mapping</code>	F	Foundation entities (3)
<code>repositories/...alignment</code> , <code>...alignment-sp</code> , <code>...indicator-mapping</code>	F	Foundation repositories (3)
<code>handlers/{capacity-sharing, innovation-development, knowledge-product, policy-change, noop}.handler.ts</code>	F	Per-indicator-type contribution handlers (Strategy pattern) + interface
<code>dto/{upsert-indicator-mapping, review-decision, list-indicators-query, bilateral-science-programs.response}.dto.ts</code>	F	Foundation DTOs

4. Functions & business logic

4.1 `TocIntegrationService` NEW

Method	Behaviour
<code>getTocResults(sp, level): Promise<TocResult[]></code>	GET <code>{HOST}/api/toc-integration/toc/results/category/{level}/initiative/{sp}</code> . Cache hit → return; miss → fetch + cache (TTL 5 min). Upstream failure: warm entry → serve stale + <code>LoggerUtil.warn</code> ; cold → <code>ServiceUnavailableException</code> (503). <code>{"response": []}</code> cached as valid empty.
<code>getTocResultsForSps(sps, levels)</code>	Parallel fan-out ($\leq Sps \times levels $ calls via <code>Promise.all</code>) → <code>Map<`\${sp}:\${level}`, TocResult[]></code>
<code>cacheKey / assertHost</code> (private)	Key builder; missing <code>ARI_TOC_INTEGRATION_HOST</code> ⇒ 503 (mirrors <code>PrmsTocService</code>)

4.2 `toc-level-rules.util` NEW — the single source of truth (D-V2-3)

`resolveResultTypeKey(indicatorId)` maps `results.indicator_id` ($\in$ `IndicatorsEnum`) → canonical key via an exhaustive record (compile-time drift protection). Unknown/null ⇒ `'unknown'` .

```
1 → capacity_sharing
2 → innovation_dev
3 → knowledge_product
4 → policy_change
5 → oicr
6 → innovation_use
```

`allowedLevelsFor(key)` returns a fresh array (callers can't mutate the source). Types with no row ⇒ `[]` ⇒ `catalogs: []` + zero upstream calls.

```
capacity_sharing → [OUTPUT]
innovation_dev    → [OUTPUT]
policy_change     → [OUTCOME, EOI]
(anything else)  → []
```

4.3 `BilateralService` — key methods MOD

Method	Role
<code>getHlosIndicatorsForResult()</code>	Read. Context 404 → rules util (type+levels) → <code>version_locked</code> → SP chain → <code>getTocResultsForSps</code> → wire mapping. Empty levels / unmapped ⇒ <code>catalogs: []</code> , zero upstream calls.
<code>toWireTocResult / toWireTocIndicator</code> (priv.)	Upstream → frozen wire shape: <code>wp_short_name=aow_code</code> (null at EOI), <code>unit_messurament=unit_of_measurement</code> , <code>type_value</code> passthrough, nulls → <code>''</code>
<code>resolveLiveTargetValue()</code> (priv.)	Shared by read & write snapshots: picks the <code>targets[]</code> entry for 2026 → <code>(target_value, 2026)</code> , else <code>(null, 2026)</code>
<code>deriveSciencePrograms()</code> (priv.)	Confirmed, level-1, prefix "SP", active-portfolio SPs from the CLARISA project; deduped, sorted
<code>getAlignment()</code>	Read-back. + <code>version_locked</code> + snapshot-sourced <code>toc_alignments[]</code> (zero upstream); ineligible ⇒ <code>[]</code>
<code>updateAlignment()</code>	Write (single transaction). See 4.4.
<code>validateTocAlignments()</code> (priv.)	Gate + structural + catalog validation; runs <i>before</i> the transaction
<code>upsertContribution / deleteContribution / reviewDecision / listIndicators</code>	Foundation indicator-mapping + review flows (dispatch to type handlers)

4.4 Write path — `updateAlignment` (design §6.3)

Only runs the ToC logic when `toc_alignments` is present; legacy bodies behave byte-identically.

- Existing gates: result exists; not PRMS-sourced (409); pool-funding contributor (400); not synced (409); `unknown_sp_codes` (400).
- Version gate** — `report_year_id ≠ 2026` ⇒ 409 { `errors.code: 'toc_mapping_version_locked'` } .
- Structural + catalog validation** — collect ALL errors into one atomic 400 { `errors.toc_alignments: [{sp_code, field, error}]` } . Six codes: `duplicate_sp_code`, `sp_not_selected`, `missing_required_fields`, `level_not_allowed`, `unknown_toc_result_id`, `unknown_indicator_id` . Cold-cache during validation ⇒ 503, nothing persisted.

4. **Per-SP upsert** (`upsertForSp` , in-transaction): "Yes" copies snapshots from the validated catalog entry; "No" nulls ToC/snapshot columns. SPs absent from the body are untouched.
5. **Cascade** (`deactivateForSps`): active ToC rows whose SP left the effective `sp_codes` are soft-deactivated — runs on *every* PATCH (R-BIL-093).
6. Side effects unchanged: audit fields, `result_review_history` entry, socket `POOL_FUNDING_ALIGNMENT_CHANGED` (payload unchanged).
7. Response = `getAlignment()` read-back ⇒ **PATCH response** ≡ **GET**.

4.5 Indicator-type handlers FOUNDATION — Strategy pattern

Interface `BilateralIndicatorTypeHandler` { `indicatorType` , `validate(dto)` , `upsert(ctx, dto)` , `delete(ctx)` }. Five implementations dispatched by `indicator_type` : `capacity_sharing`, `innovation_development`, `knowledge_product`, `policy_change`, and `N00P` (fallback).

5. Migrations

Append-only. The bilateral epic added migrations `1779190000001` ... `1779190000015` . The ToC table is the only schema change from ToC-mapping-v2; the rest are foundation.

Migration	Creates / changes	Tag
<code>...0001</code>	<code>agresso_contract.is_pool_funding_contributor</code> tag column	F
<code>...0003 / 0004</code>	New bilateral result statuses + status workflow	F
<code>...0005</code>	<code>ReportingPlatform</code> row (bilateral)	F
<code>...0006 / 0007</code>	<code>result_pool_funding_alignment</code> + <code>..._sp</code>	F
<code>...0008</code>	<code>result_pool_funding_indicator_mapping</code> (+5 FKs)	F
<code>...0009</code>	<code>result_review_history</code>	F
<code>...0011</code>	<code>bilateral_project_mapping</code> (generated-col partial-unique)	F
<code>...0014</code>	Fix alignment partial-unique (generated column)	F
<code>...0015</code>	<code>result_pool_funding_toc_alignment</code> + generated col + unique + FK	NEW

Verification (T-05): applied live on the dev DB; unique index proven via `ER_DUP_ENTRY` on a second active insert; reverted & re-applied cleanly. `down()` drops the FK then the table.

6. API surface

All under `/api/v1/results/:resultCode/pool-funding-alignment` , controller guarded by `RolesGuard` ; mutations add `ResultOwnerGuard` .

Method	Path	Purpose	Tag
GET	<code>/</code>	Alignment read-back (+ <code>version_locked</code> , <code>toc_alignments[]</code>)	MOD
GET	<code>/science-programs</code>	SP picker for the mapped project	F
GET	<code>/hlos-indicators</code>	Per-SP × per-level ToC catalogs (frozen envelope)	MOD
GET	<code>/indicators</code>	Indicator-mapping list	F
PATCH	<code>/</code>	Update alignment (+ <code>toc_alignments[]</code> , gate, cascade)	MOD
POST/PATCH/DELETE	<code>/indicators/:code/contribution</code>	Typed contribution CRUD (handler dispatch)	F

Errors: 404 result not found · 503 cold-cache upstream · 400 atomic per-alignment validation · 409 `toc_mapping_version_locked` / PRMS-sourced / synced. All via the `ServerResponseDto` envelope. Swagger renders the frozen response schema (typed `@ApiResponse` 200/404/503 on the catalog read).

7. Tests

109 unit tests across the two modules (all green at last run; full server suite 284 suites / 1660 tests; coverage 80.2 / 71.5 / 80.9 / 80.0 vs the 60% floor).

Spec file	Tests	Covers
<code>bilateral.service.updateAlignment.tocAlignments.spec.ts</code>	24	Write path: per-SP independence, "No" rows, cascade, all six 400 codes (atomic), 409 gate, legacy bypass, validation-time 503
<code>bilateral.controller.spec.ts</code>	13	Delegation, Swagger metadata, real <code>RolesGuard</code> allow/deny
<code>bilateral.service.spec.ts</code>	15	Read-back mapping, drift (snapshot survival), <code>version_locked</code>
<code>bilateral.service.getHlosIndicatorsForResult.spec.ts</code>	8	Catalog read: mapped/unmapped/empty/ <code>allowed_levels: []</code> , target resolution (handoff-parity fixtures)
<code>bilateral.service.getScienceProgramsForResult.spec.ts</code>	8	SP picker chain
<code>toc-integration.service.spec.ts</code>	12	Cache TTL, warm-stale, cold-503, empty payload, fan-out count, missing host
<code>toc-level-rules.util.spec.ts</code>	12	Rule table rows, enum mapping, unknown/null, immutability, 2026 constant
<code>result-pool-funding-toc-alignment.repository.spec.ts</code>	9	find/order, upsert insert & update-in-place, "No" nulls, EntityManager routing, deactivate
<code>bilateral.service.normalizeLeverCodes / sourceReadOnlyGate.spec.ts</code>	4 + 4	Lever normalization; PRMS source read-only gate

8. Integrations & configuration

Item	Detail
lambda-toc (new upstream)	<code>ARI_TOC_INTEGRATION_HOST</code> = <code>https://lambda-toc.clarisa.cgiar.org</code> . No auth. Server-side only (kept behind JWT). Missing host ⇒ 503.
Active portfolio	<code>ARI_BILATERAL_ACTIVE_PORTFOLIO</code> (default <code>P25</code>) — filters Confirmed SPs in the resolution chain
CLARISA	Projects + science-programs + cgiar-entities (SP resolution); unchanged
Socket.IO	<code>POOL_FUNDING_ALIGNMENT_CHANGED</code> — payload unchanged (FE refetches on event)
Resilience	5-min in-memory cache per <code>(sp, level)</code> ; warm-stale serve on upstream blip; cold-cache 503; observability via <code>LoggerUtil</code> warn/error

Operational caveat (RB-1, open): `lambda-toc.clarisa.cgiar.org` did not resolve on the office DNS (NXDOMAIN); resolves via 8.8.8.8 → `3.90.182.187` . The testing/prod servers must be able to resolve the host, or the catalog read returns a cold-cache 503. Local fix: a per-domain resolver (`/etc/resolver/clarisa.cgiar.org`).

9. Decisions & open follow-ups

Key design decisions

- D-V2-3** — the level rule lives only in `toc-level-rules.util` , shared by read + write (no FE re-derivation, no drift).
- D-V2-4** — snapshot column keeps the upstream spelling `unit_messurament` ; the rename to `unit_of_measurement` is a wire-layer concern.
- D-V2-7** — version gate anchors on `results.report_year_id` vs the hardcoded constant `MAPPABLE_LIVE_VERSION = 2026` .
- D-V2-8** — validation is atomic: the whole PATCH 400s with all errors at once; no partial persist.
- R-BIL-093** — the SP-deselection cascade runs on every PATCH (incl. legacy bodies); only the version gate is `toc_alignments` -scoped.

Open follow-ups

- T-10 (gated)** — delete the legacy PRMS pair path (`src/domain/tools/prms-toc/` , `ARI_PRMS_TOC_HOST` , dead AOW fan-out) once cutover is verified. Still present, compilable-but-unused.

- **RB-1** — lambda-toc DNS on testing/prod (see §8).
- **RB-4** — relay the frozen read-back + error contracts to the STAR FE.
- **OQ-V2-2/3/5/6** — BA decisions (indicator-type filter; one-alignment-per-SP cardinality; level rules for other result types; target year 2026 vs 2025). Build shipped on recorded assumptions.

Bilateral Module — Technical Documentation · generated 2026-06-17 from `server/researchindicators` on branch `AC-1594-bilateral-module-v2` . Full decision & execution trail preserved at `docs/specs/archive/2026-06-17-bilateral-module/` and `...-toc-mapping-v2/` .